

AI Maze Escape

Intelligent Pathfinding & Adversarial AI

PYTHON + PYGAME

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What is AI Maze Escape?

Interactive Maze

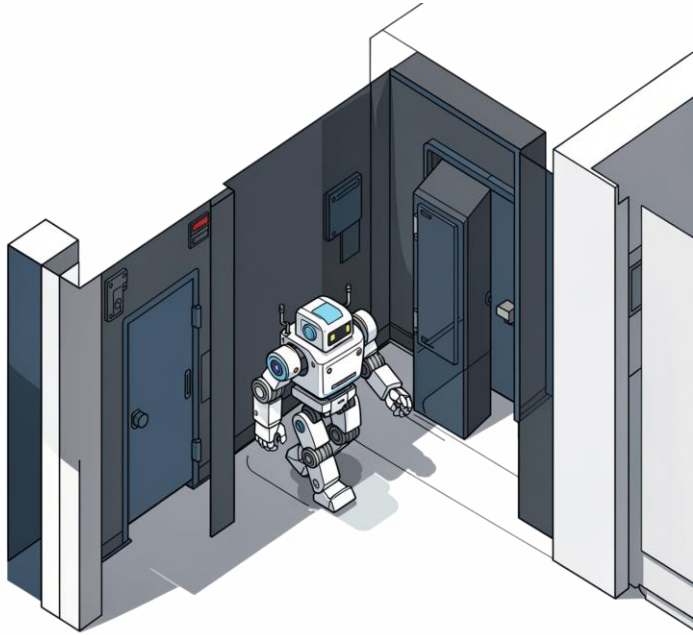
Agents navigate a changing maze.

AI Pathfinding

Finds efficient routes.

Enemy Pursuit

Adversarial AI chases the player.



The Challenge



Goal

Navigate from **Start** → **Exit**



Obstacles

Walls and maze structure block the path



Constraint

Intelligent enemy actively pursues the player



Outcome

Win by reaching exit — lose if caught

Project Objectives

01

Maze Environment

Build an interactive, procedurally generated maze

02

Pathfinding Algorithm

Implement intelligent A* search for navigation

03

Adversarial AI

Develop Minimax-driven enemy agent

04

Visualization

Display algorithm decisions in real time

05

Playable Game

Deliver an engaging, complete game experience

A* Search — Pathfinding

 $f(n) = g(n) + h(n)$

$g(n)$

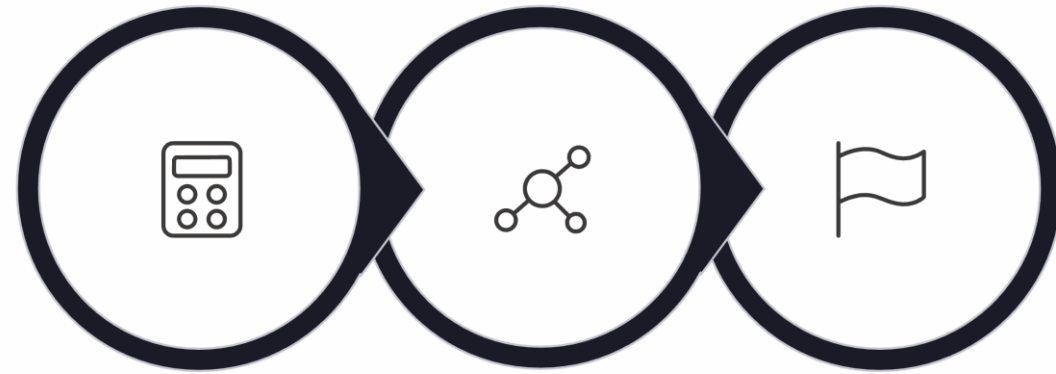
Actual cost from start node

$h(n)$

Manhattan Distance heuristic to goal

$f(n)$

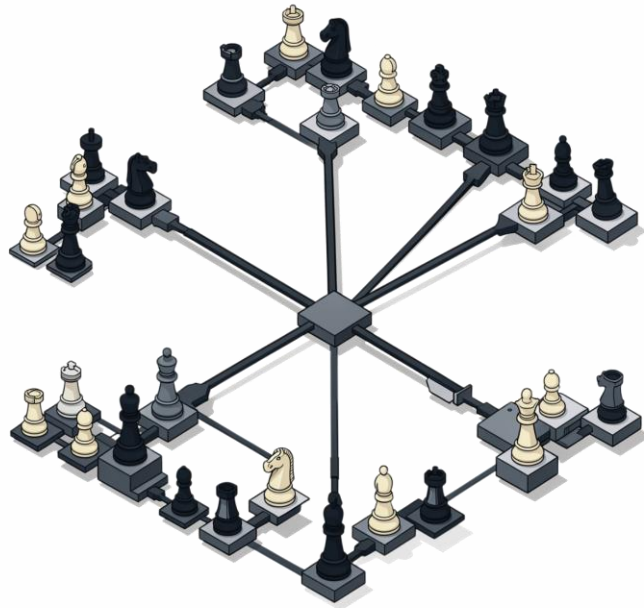
Total estimated path cost



Evaluate $f(n)$

Expand
Lowest-Cost

Reach Optimal
Path



Minimax — Enemy AI

Purpose: Intelligent adversarial decision-making

→ Evaluates future moves

Explores the game tree with
lookahead depth

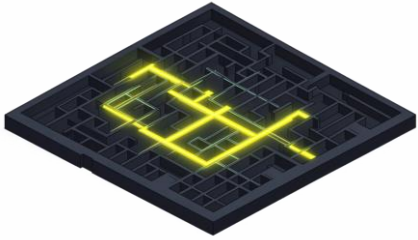
→ Best move selected

Explores → evaluates outcomes → commits to optimal move

→ Opposite goals

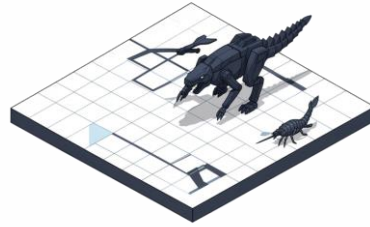
Enemy maximizes capture; player
minimizes risk

How They Work Together



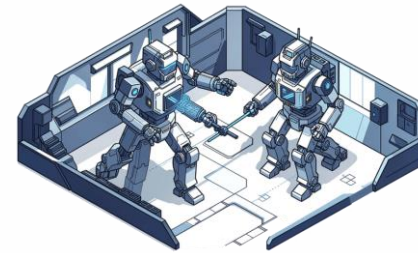
A* Search — Player

Computes the shortest safe route to the exit in real time



Minimax — Enemy

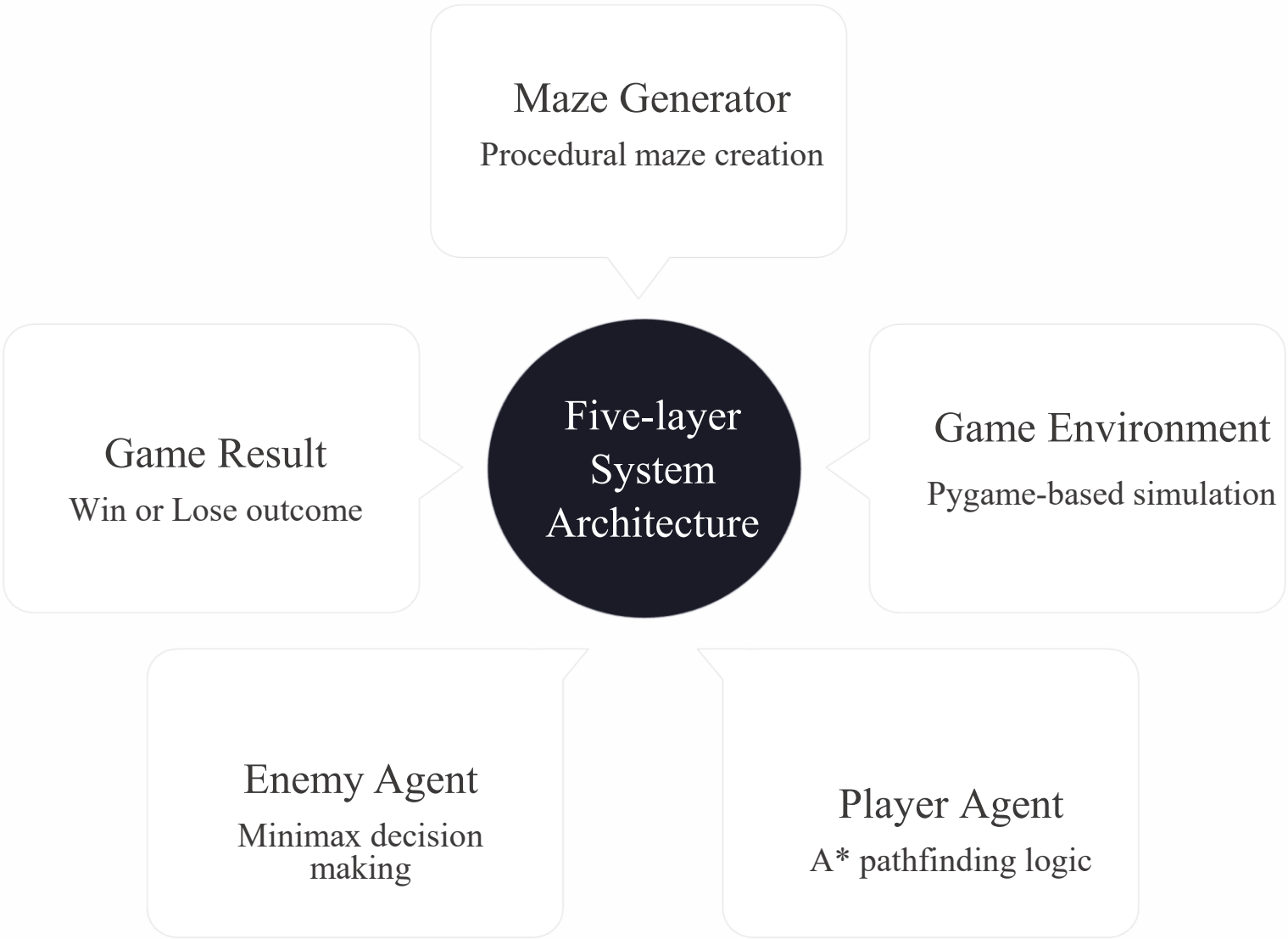
Simulates future states to intercept and corner the player



Dynamic Interaction

Two AI algorithms solving different problems in the same world — creating emergent gameplay

System Architecture



Game Features



Procedural Maze

Auto-generated levels & collision detection



A* Visualization

Live pathfinding rendered on screen



Minimax Enemy

Real-time adversarial pursuit behavior



Score & Timer

Win/lose conditions with scoring system



Results & Future Work

Expected Results

Optimal
navigation with A*

Intelligent
adversarial enemy

Clear real-time AI
visualization

Playable, engaging
experience

Future Improvements

Alpha-Beta
pruning for faster
Minimax

Multiple
coordinated
enemies

Dynamic difficulty
levels

RBFS & IDA*
alternatives

Thank You!